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COLLEGE OF ENGINEERING
Akurdi, Pune

Presentation
On
Subject: UI and UX Design
by

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Unit 1: Introduction to UI and UX

Session-1

Overview of system design with a focus on UI/UX.
Design Thinking Process

Learning Outcomes of the session:

After this session, students will be able to:

- . Explain the concept of system design in software engineering**
- . Describe the role of UI/UX in overall system success**
- . Identify real-world examples where UI/UX impacts user adoption**

Content:

Content	Learning / Methodology	Faculty Approach	Typical Student Activity	Skill/ Competency Developed
Introduction: Overview of system design with a focus on UI/UX. Design Thinking Process: Empathize, Define, Ideate, Prototype, Test, Human-centered design principles, Difference between UI and UX, Role of UX in system-level architecture.	PPT, Experimentation	Explain	Understand	Critical Thinking - Observation

Introduction to System Design and UI/UX

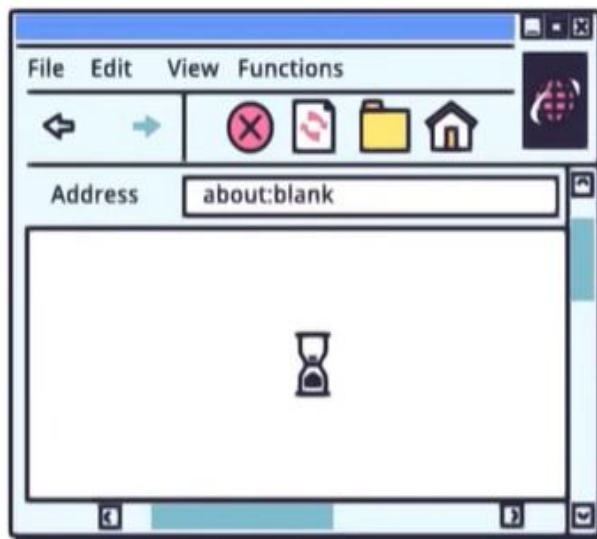
UI & UX DESIGN

Syllabus





Sam
(College Student)



- It was hard to read
- Buttons didn't work properly
- Colors hurt Sam's eyes



Sam's Friend



Sam
(College Student)



Let's go on a journey with Sam to find out what
UI/UX is and why it's so important!



Sam's Friend

What is UI?

UI (User Interface) is **everything a user sees and interacts with** on a screen.

It includes:

- Buttons
- Icons
- Text
- Forms
- Colors
- Layouts

UI is the visual part of a system that allows users to interact with it.

System

UI Example

ATM

Buttons, screen text

Mobile App

Icons, menus

Website

Navigation bar, forms

College Portal

Login screen, dashboard

If you can **see it or touch it**, it is part of UI.

Role of UI

UI helps users to:

- Understand options
- Perform actions easily
- Navigate the system
- Avoid confusion

Bad UI → Confusion

Good UI → Ease of use

What is UX?

UX (User Experience) is the **overall feeling and experience** a user has while using a system. It includes:

- Ease of use
- Speed
- Comfort
- Satisfaction
- Emotions

UX is how users feel while using a product.

UX Examples (Very Effective)

Example 1: Railway Ticket Booking

- Easy booking → Good UX
- Server crash at payment → Bad UX

Example 2: College Exam Portal

- Clear instructions → Good UX
- Confusing navigation → Bad UX

UI

Visual
design

Screen elements

How it looks

Part of system

UX

User experience

User journey

How it works

End-to-end experience

UI is a subset of UX.

What is UI/UX?

UI → User Interface



You know exactly where to press or click without even thinking about it.

UX → User Experience



Does it make your life easier or harder?



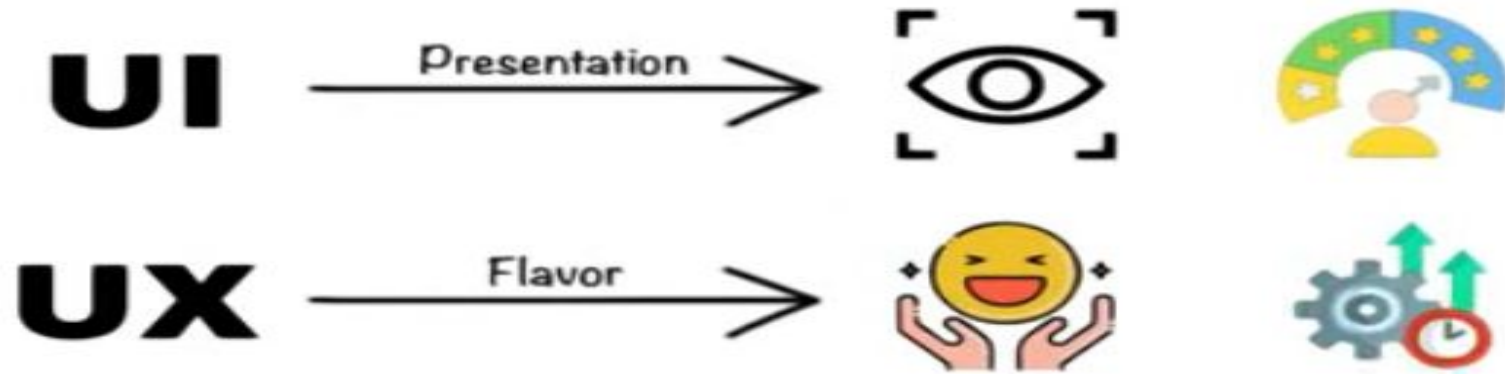
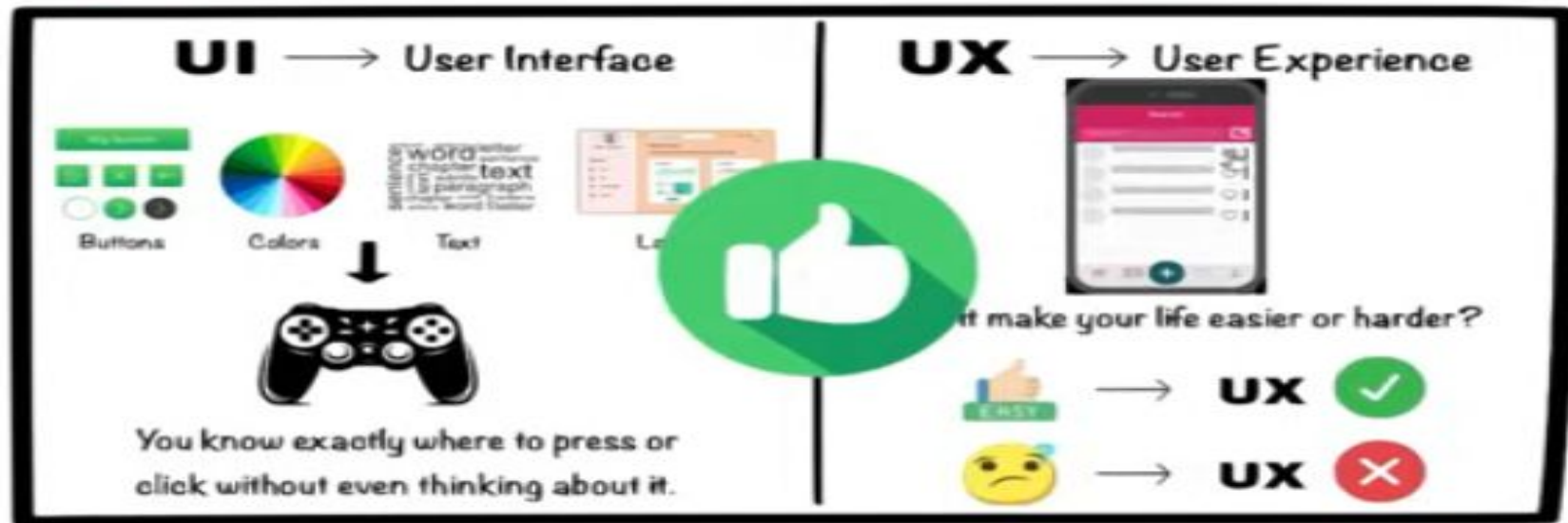
UX



UX



What is UI/UX?





- ▶ The buttons were too small to tap.
- ▶ Nobody could figure out where to add assignments.
- ▶ The colors were so bright that it hurt their eyes.
- ▶ The app felt "messy" – like too much was happening on one screen.

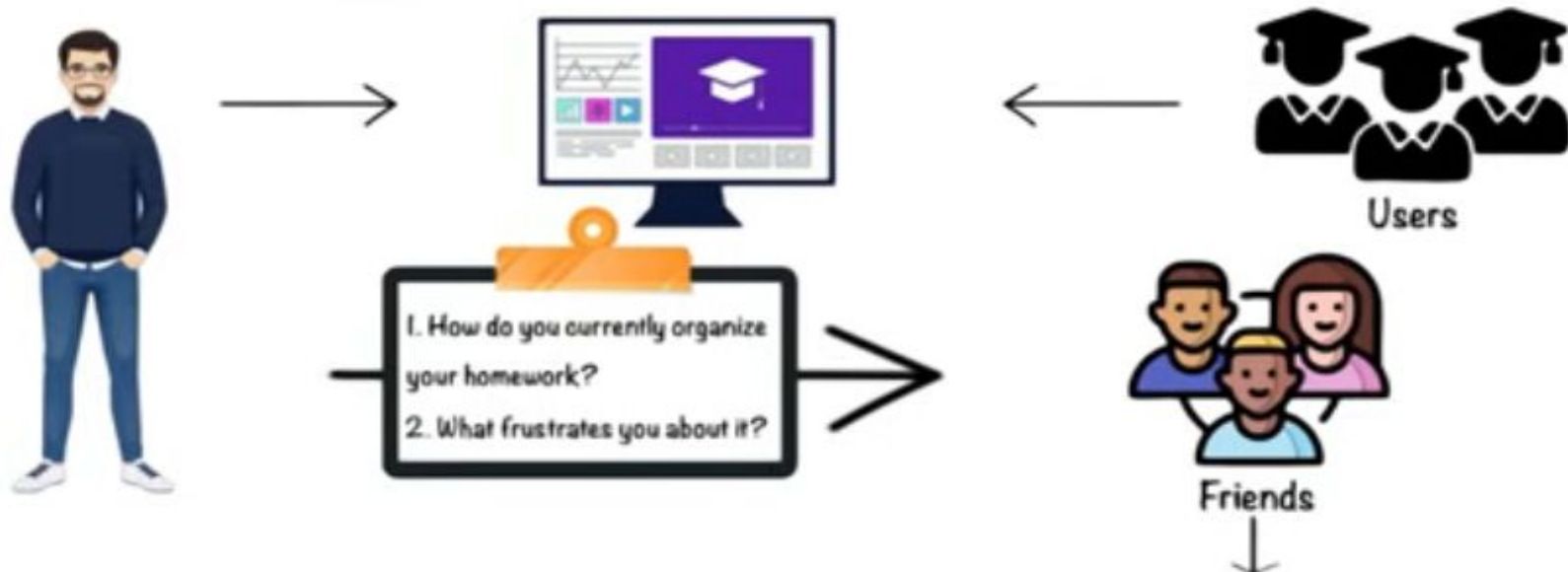


Sam



How Sam Solved It with UI/UX?

I. Understanding the User (User Research)



→ Surveys and Collect Feedback

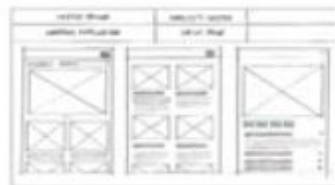
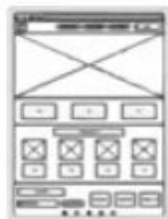


→ Must-Have Features and Common Issues



How Sam Solved It with UI/UX?

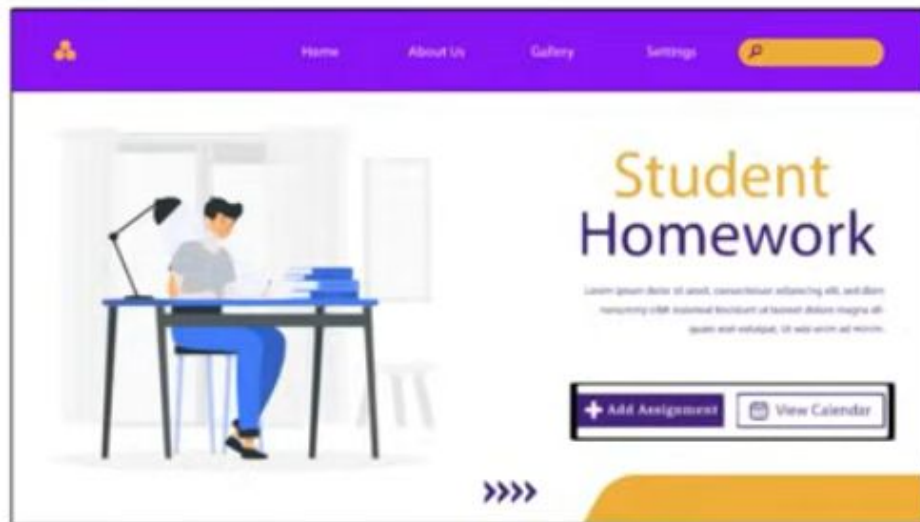
2. Sketching the Plan (Wireframing)



This step helped Sam focus on the app's structure and how users would interact with it.

How Sam Solved It with UI/UX?

3. Choosing the Right Look (Visual Design)



A_A



Canva



Visual styles and
Create mood boards

COOLORS



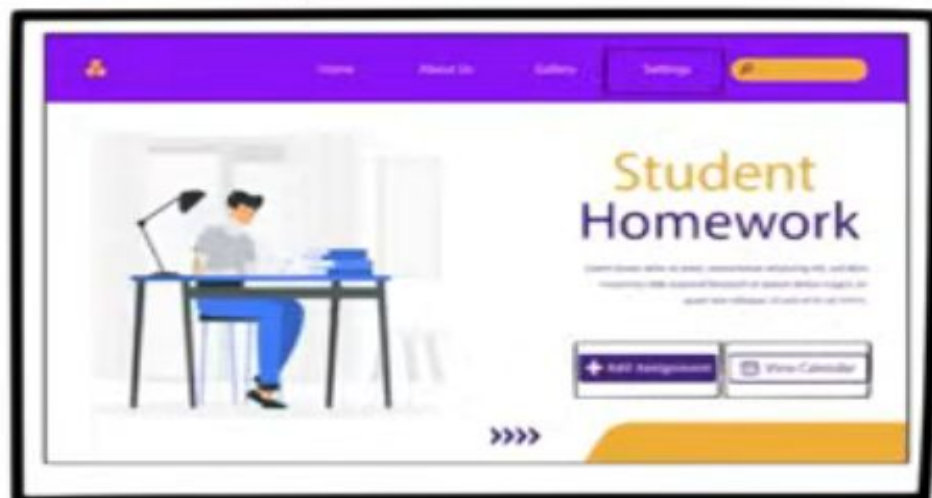
Generate harmonious
Color schemes

This step ensured the app looked professional and consistent while being pleasant to use.



How Sam Solved It with UI/UX?

4. Organizing the Screens (Information Architecture)



This organization made it easier for users to find what they were looking for without feeling overwhelmed.

How Sam Solved It with UI/UX?

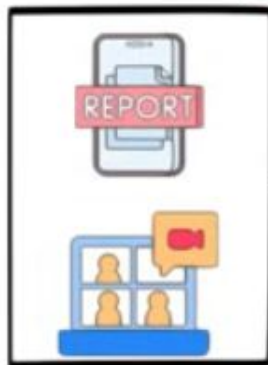
5. Testing and Improving (Usability Testing)



Real-world Use



1. Save Button Bigger
2. Color-coding Feature





Creating wireframes, designing thoughtfully, and testing the app,
Sam turned their idea into something truly useful.



It helps solve real problems and creates designs
that people love to use!

Wireframe

A wireframe is a basic, skeletal blueprint for a website, app, or digital product, focusing on layout, structure, and functionality rather than visual design (colors, fonts)

Miro :

Used for Project Management , Design and planning a project.

Responsibility Comparision

UI Designer

Collaborating with product management and engineering to define and implement innovative solutions.

Executing all interaction design stages from concept to final hand-off to engineering.

Conceptualizing original ideas that bring simplicity and user friendliness to complex design roadblocks.

UI/UX Designer

Gathering and evaluating user requirements, in collaboration with product managers and engineers.

Illustrating design ideas using storyboards, process flows and sitemaps.

Designing graphic user interface elements, like menus, tabs and widgets.

UX Designer

Conducting user research and testing.

Developing wireframes and task flows based on user needs.

Collaborating with designers and developers to create intuitive, user-friendly software.

What is System Design?

System Design is the process of defining:

- System components
- Interactions between components
- Data flow
- User interaction with the system

System design is how we plan **what a system does and how users interact with it.**

- In system design, UI/UX acts as a bridge between technical functionality and human needs. A well-designed system may fail if users find it confusing or difficult to operate. For example, a banking application may have strong security and fast performance, but if users cannot easily find the “Transfer Money” option, the system fails in usability. Hence, UI/UX is considered a core component of system-level design rather than an optional add-on.
- UI/UX also impacts business success. Good design improves user engagement, reduces errors, increases customer satisfaction, and lowers training costs. Companies like Google, Apple, and Amazon invest heavily in UI/UX because it directly affects user retention and brand value.
- In conclusion, system design with a focus on UI/UX ensures that technical solutions are aligned with human needs. It integrates aesthetics, functionality, and usability into a unified system, making software efficient, accessible, and user-friendly.

Where UI/UX Fits in System Design

Traditional system design focused mainly on:

- Backend logic
- Databases
- Performance

Modern system design also focuses on:

- User experience
- Usability
- Accessibility
- Interaction flow

Example:

A banking system may work perfectly technically, but if users cannot find “**Transfer Money**”, the system fails.

Design Thinking Process

Design thinking is an iterative process in which we seek to understand the user, challenge assumptions, and redefine problems in an attempt to identify alternative strategies and solutions that might not be instantly apparent with our initial level of understanding.

Design thinking is a structured method to solve problems by understanding users first.

There are many variants of the design thinking process in use today, and they have from three to seven phases, stages, or modes.

Five Stages of Design Thinking

1. Empathize – with your users
2. Define – your users' needs, their problem, and your insights
3. Ideate – by challenging assumptions and creating ideas for innovative solutions
4. Prototype – to start creating solutions
5. Test – solutions

1. Empathize

Empathize means understanding the users, their needs, problems, and emotions by observing and interacting with them.

This stage focuses on understanding users and their problems. Designers observe user behavior, conduct interviews, surveys, and usability studies to gain insights into user needs and pain points.

Example:

In a hospital management system, designers observe how nurses record patient data. They may find that existing software is time-consuming and causes data entry errors.

Purpose:

To understand what users actually need, not what we assume they need.

Example:

In a **hospital management system**, designers observe nurses and doctors while they enter patient data.

Scenario:

Nurses complain that entering patient details takes too much time during emergencies. Designers note their struggles, workload, and environment.

2. Define

The **Define** stage focuses on clearly identifying the problem based on user observations. In this stage, the collected data is analyzed to clearly define the problem. A well-defined problem statement guides the entire design process. The **Define** stage focuses on clearly identifying the problem based on user observations.

Purpose:

To convert user observations into a clear problem statement.

Example:

“Hospital staff need a faster and simpler way to enter patient details during emergencies.”

Scenario:

After observing delays, designers define the problem as:

“Too many steps in data entry are causing delays in patient treatment.”

“Users need a faster and simpler way to enter patient data during emergencies without navigating multiple screens.”

3. Ideate

Here, designers brainstorm multiple solutions without limitations. Creativity is encouraged, and no idea is rejected initially. Ideate means generating multiple ideas and solutions for the defined problem.

Purpose:

To think creatively and explore different solutions.

Example:

Ideas for hospital system:

1. Voice-based data entry
2. One-click emergency form
3. Pre-filled patient records
4. Touch-based shortcuts

Scenario:

A team brainstorms and sketches different interfaces that reduce typing and clicks.

4. Prototype

A **prototype** is a simple working model of the solution.

Purpose:

To visualize ideas before final development.

Example:

A clickable screen showing:

- Emergency button
- Auto-filled patient details
- Large icons for easy use

Scenario:

Doctors test a sample screen on a tablet before actual software development begins.

5. Test

Testing involves checking the prototype with real users and collecting feedback.

Purpose:

To identify errors and improve usability.

Example:

Doctors use the prototype and report:

- Buttons are too small
- Some steps are unnecessary

Scenario:

Nurses test the prototype and suggest larger buttons and fewer steps.

Based on feedback, the design is improved and retested until it works smoothly.

What is human-centered design?

Human-centred Design is a design approach that prioritizes the needs, wants, and perspectives of people in the design process.

The Goal

- The goal of **Human-centred Design** is to create products, services, and experiences that are meaningful, relevant, and effective for the people who will use them.
- It starts with a deep understanding of the people who will use the product or service, and involves co-design and co-creation with those users. This ensures that the design solution meets the needs and expectations of the end-users.

Human-Centered Design:

1. (HCD) is a design approach that focuses on creating systems, products, and services by deeply understanding the needs, behaviors, limitations, and expectations of users. Instead of designing based only on technical feasibility, HCD ensures that the system is comfortable, efficient, and meaningful for real users. It places humans at the center of the design process and emphasizes usability, accessibility, and user satisfaction.
2. HCD is widely used in software development, healthcare systems, education platforms, mobile applications, and product design. The goal is to create solutions that are easy to use, reduce user effort, and improve overall experience. This approach also reduces errors, increases adoption rate, and improves productivity.

Key Principles:

1. **Usability**

Systems should be easy to learn and use.

Example: A mobile banking app with clear icons and simple navigation.

2. **Accessibility**

Systems should be usable by people with disabilities.

Example: Screen reader support and high-contrast modes.

3. **Consistency**

Similar actions should behave similarly across the system.

Example: Same button style for “Submit” across all pages.

4. **Feedback**

Users should know what action has occurred.

Example: Confirmation message after form submission.

5. Error Prevention and Recovery

Systems should prevent errors and allow easy recovery.

Example: “Undo” option or warning before deleting data.

6. User Control

Users should feel in control, not forced by the system.

Example: Ability to cancel or edit actions.

Human-centered design improves satisfaction, reduces training time, and increases system adoption. It is widely used in healthcare systems, educational platforms, and mobile applications.

Difference between UI Design & UX Design



UI Design	Aspect	UX Design
Visual elements of a product	Definition	User's overall experience with a product
Create attractive, intuitive interfaces	Objective	Improve usability and satisfaction
Adobe XD, Sketch, Figma	Tools	Balsamiq, Axure, Figma
Layout, colors, typography	Focus Area	Usability, functionality, navigation
Visually engaging interface	Outcome	Seamless user experience
Design of a login screen	Example	Streamlining login process

The Difference between

UX

Interaction Design

Wireframes

Information Architect

User Research

Senarios



UI

Visual Design

Colours

Graphic Design

Layouts

Typography

EXAMPLE 1: COLLEGE EVENT REGISTRATION PORTAL

UI

- Registration form design
- Submit button Arrangement
- Event cards and icons

UX

- How easily students find event details
- Whether reminders are sent
- Confirmation after registration

Teaching Line:

Nice-looking form (UI) but no confirmation message → bad UX.

Example

UI Focus

UX Focus

Banking App

Buttons, colors

Ease of transfer

Event Portal

Form layout

Registration experience

ATM

Screen & buttons

Task completion

Shopping App

Images & icons

Checkout flow

Exam Portal

Screen layout

Stress-free usage

Role of UX in System-Level Architecture

- The role of User Experience (UX) in system-level architecture is to bridge the gap between technical infrastructure and user needs, ensuring that backend, data, and service structures create a seamless, intuitive, and efficient experience for the end-user. While technical architects focus on stability, scalability, and performance, UX architects (or Systems UX researchers) ensure these technical choices support user workflows and do not introduce unnecessary friction

SIMPLE EXAMPLE 1: ONLINE EXAM SYSTEM

Technical (System-Level) View

- **Backend handles:**
 - Question loading
 - Timer management
 - Auto-submit logic
 - Database storage
- **System is:**
 - Fast
 - Scalable
 - Secure

Example

UX View

- Students need:
Clear timer visibility
Auto-save answers
Warning before submission
Smooth navigation

UX Bridging the Gap

- UX suggests:
Backend auto-save every 10 seconds
Warning popup before time ends
Progress indicator

Result:

Even if backend is strong, UX ensures students feel confident and stress-free.

Key Aspects of UX in System-Level Architecture

- **Information Architecture (IA) as the Foundation:** IA defines how content and data are organized, structured, and labeled to ensure user-friendly navigation and search. Eg :College Website- organized content logically
- **Translating User Needs into Technical Constraints:** UX professionals identify user requirements and translate them into functional, non-functional, and data requirements for developers, such as defining user flows, data models, and interaction models Eg:- Event Registration system/College Website
- **Systems Thinking and Human-Technical Interoperability:** Systems UX researchers focus on the entire ecosystem, ensuring that human-computer interactions are smooth, especially when dealing with complex, multi-service systems (e.g., AI integration, microservices). Eg: College Website - about event registration confirmation.
- **Balancing Technical and Design Goals:** UX works with architects to ensure that technical decisions (e.g., opting for microservices over a monolith) still allow for a consistent, coherent user experience across the entire product. Eg:College ERP-performance scalability, API- System should Technically strong and easy to use

Key Contributions to Architecture Decisions

- **Performance & Responsiveness:** UX advocates for speed (e.g., keeping delays under 100ms) by influencing how data is fetched and rendered.
- **Scalability & Reliability:** UX ensures that as systems scale (e.g., adding more backend services), the interface remains intuitive and consistent.
- **Graceful Handling of System Failures:** UX designers plan for edge cases and system errors (e.g., offline modes, API failures), ensuring user trust is maintained even during technical interruptions.
- **Enterprise Integration:** In complex B2B systems, UX ensures that the software, which is often heavily integrated with existing, outdated infrastructure, remains usable and efficient for workers.

Techniques to improve the UI UX contribution

Key Deliverables & Techniques

- **Sitemaps and Process Flows:** Mapping out the structure of the system.
- **System/User Journey Maps:** Mapping out the user's experience across multiple touchpoints.
- **User Personas & Role Definition:** Aligning system capabilities with user behaviors.
- **Wireframes and Prototypes:** Visualizing interaction patterns before full-scale development.

In summary, UX at the system level is about ensuring that the technical "engine" is designed to power a great user experience rather than just a functional, yet frustrating, product.

Activity 1-UI vs UX – Design Thinking in Action

Group Formation

- Divide class into **groups of 4–5 students**
- Each group selects **one system**

Suggested Systems

- Online Exam Portal
- College ERP System
- Food Delivery App
- Banking App
- Event Registration System